PENK Project — Wet-Lab Decision Sheet

iGEM Patras · from the dry-lab modelling team · one page, four decisions

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1 · Which plasmid to use → Plasmid 3 (PENK-IRES-mEGFP)

Use	Avoid	Why
Plasmid 3 — cleanest	**Plasmid 1** — defective	Plasmid 3 has PENK cap-driven, full-length, signal peptide intact, byte-identical to reference. Plasmid 1 is missing the 7th enkephalin core (YGGFMRF truncated).

- **Plasmid 2** works but PENK sits behind P2A → loses its start Met, gains a proline → may impair secretion. Fine as backup, not first choice.
- **Plasmid 4** is OK (gene order correct) — second choice after 3.

> ⚠ **The plasmid-1 defect is invisible on a gel.** It is a point truncation. Only **Sanger sequencing of the PENK 3′ end** catches it. Do not clear plasmid 1 on gel QC alone.

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2 · Before you transfect → run the 2-digest QC gel

Digest	Enzymes	What it tells you
A — insert release	AflII + XbaI	Linker type: **~2.16 kb band = IRES**, **~1.63 kb = P2A**. Constant 5,345 bp backbone = loading control.
B — identity fingerprint	NcoI (5 cuts)	Gene order: read the **second-largest band** (see gel figure).

- Run both per prep against a **1 kb Plus ladder**. Match to the virtual gel in the protocol.
- Full recipe + band table: `restriction_qc_protocol.md`, `restriction_qc_gel.png`.
- **Then Sanger-sequence** the PENK ORF (mandatory for plasmid 1; recommended for all).

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3 · Order these siRNAs (NEP / MME knockdown arm)

All target the CDS shared by **all four MME variants** → one duplex silences every isoform. Standard **3′ dTdT** overhangs; order as annealed duplexes.

Priority	Name	Guide / antisense (5'→3')
1st choice	si-MME-3 (CDS1224)	`AUAGUUUGCACAACGUCUCdTdT`
2nd	si-MME-4 (CDS1590)	`UAUCCACUCAUCUUUGUCCdTdT`
3rd	si-MME-2 (CDS1032)	`UUCUGGAGCAUAAACAACCdTdT`
4th	si-MME-1 (CDS759)	`UUCCUGACGAAUCAAUCUGdTdT`

- **Validate ≥2** by qPCR of MME at 48 h (expect ≥70% knockdown) + a NEP-activity or CD10 Western.
- Always include a **non-targeting scrambled control**.
- Priority order reflects added accessibility scoring (`sirna_design_v2.md`); si-MME-3 has perfect specificity + open target site.

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4 · How to deliver → co-formulate siRNA + plasmid in ONE lipoplex

- **Do not deliver them separately.** NEP is degraded extracellularly but silencing is cell-autonomous — the enkephalin-producing cells and the NEP-silenced cells must be the **same** cells. Co-package both nucleic acids in the same lipoplex, same locus.
- **Lipid choice:** an **ionizable lipid (MC3-type)** is preferred over permanent-cation (DOTAP) — the MD shows it binds DNA at endosomal pH but releases at physiological pH (needed for circulation + endosomal escape). DOTAP binds constitutively (harder to release).
- **Where:** local/regional delivery — **intrathecal (spinal dorsal horn)** is the primary target. Lipoplexes do not cross the blood–brain barrier.

One-line summary

Use plasmid 3, QC it by 2-digest gel + Sanger, order si-MME-3 (+ one more), and co-formulate siRNA + plasmid in an ionizable-lipid lipoplex delivered intrathecally.

Full rationale, methods, and caveats: consolidated project brief (*`project_brief.pdf`*).